

Machine Learning

Diabetes Risk Prediction using Logistic regression and Random Forest with Cross-Validation

Problem Statement:

Diabetes is a chronic disease that affects millions worldwide, and early prediction is crucial for timely medical intervention. The **PIMA Indians Diabetes Dataset**, which contains patient health records such as glucose levels, blood pressure, BMI, and insulin, provides an opportunity to apply machine learning for risk prediction.

The challenge is to develop models that can accurately predict whether a patient is likely to develop diabetes based on these health parameters.

To address this:

- **Logistic Regression** will be used as a baseline model for its simplicity and interpretability in binary classification.
- **Random Forest** will be applied as an ensemble method to capture complex, non-linear relationships between features and outcomes.
- **Cross Validation (k-fold CV)** will be employed to ensure the models are robust, reliable, and not overfitted.

This project aims to compare the performance of these models, balancing accuracy with interpretability, and demonstrate how machine learning can support healthcare decision-making.